

The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

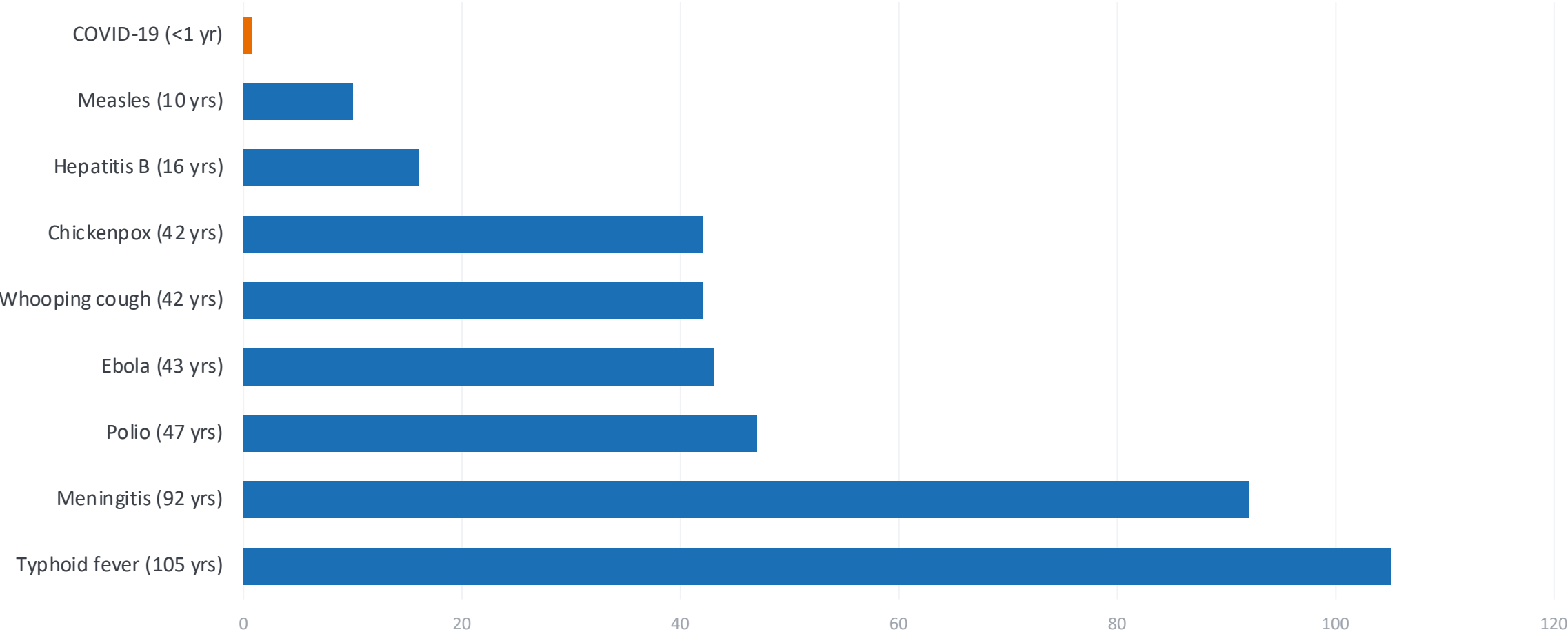
A data-driven overview of two centuries of vaccine impact | 1796 – 2020

Source: Our World in Data • ourworldindata.org/vaccination

For public health professionals and policy analysts

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Years from pathogen ID to vaccine licensure



Modern tools — mRNA technology, genome sequencing, advanced microscopy — have enabled record-breaking vaccine design.

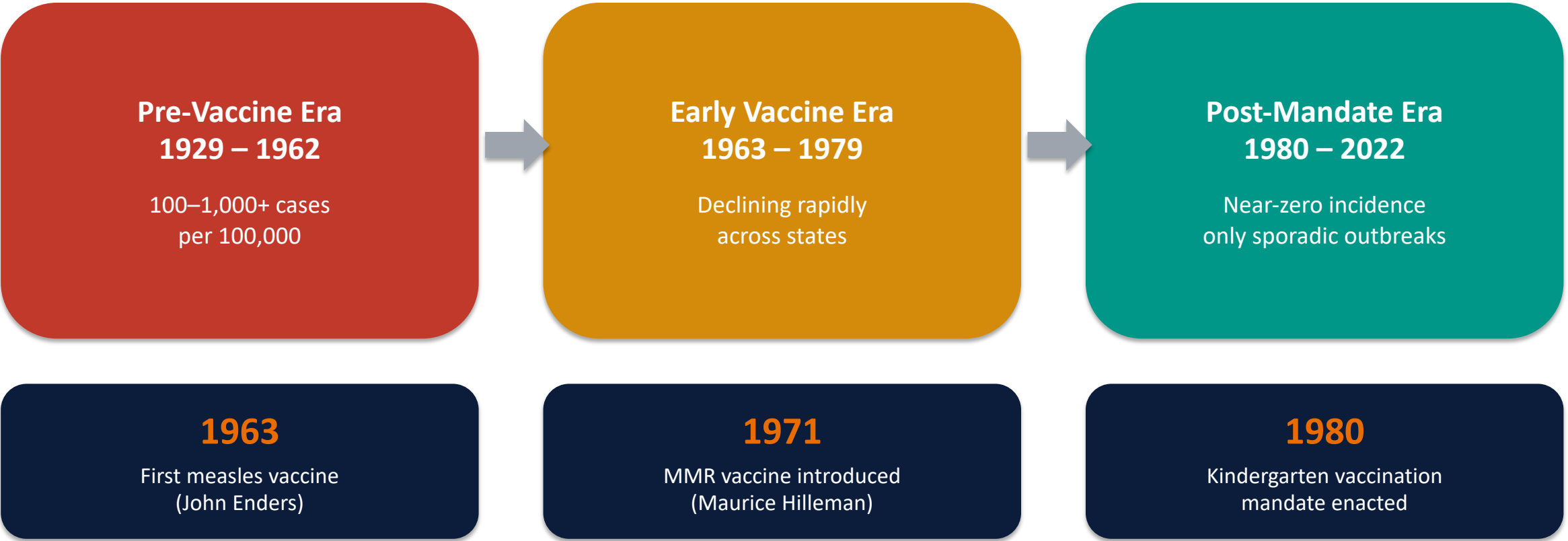
100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre-vaccine vs. post-vaccine annual incidence and mortality (per million population/year) • Source: Roush & Murphy, 2007

Disease	Pre-Vax Cases	Post-Vax Cases	Case Δ	Pre-Vax Deaths	Death Δ
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute Polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Measles incidence per 100,000 by state, 1929–2022 • Heatmap visualization



Before 1963, nearly every state experienced high measles incidence. After the 1980 mandate, cases dropped to near zero across all states.



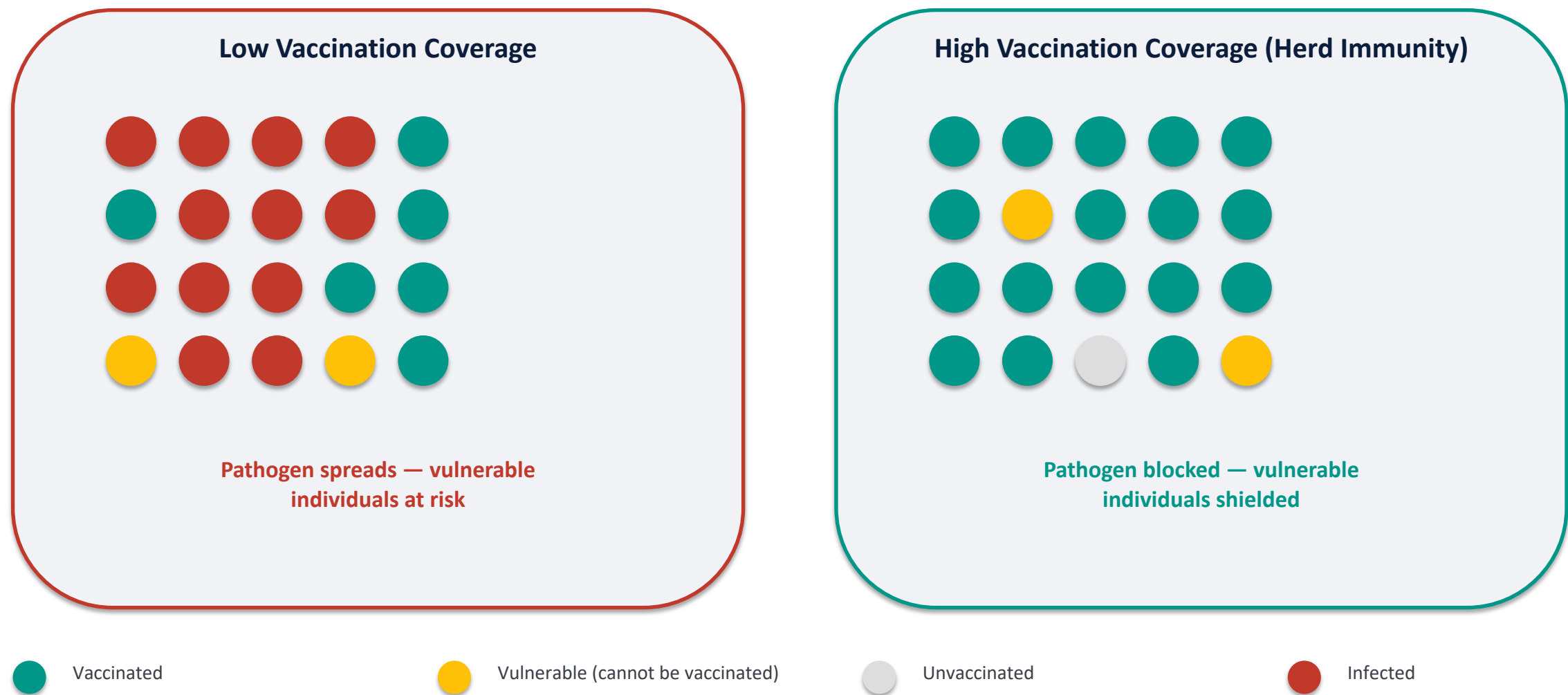
Smallpox: The Only Human Disease Eradicated Through Vaccination



Why Smallpox Was Uniquely Suited for Eradication

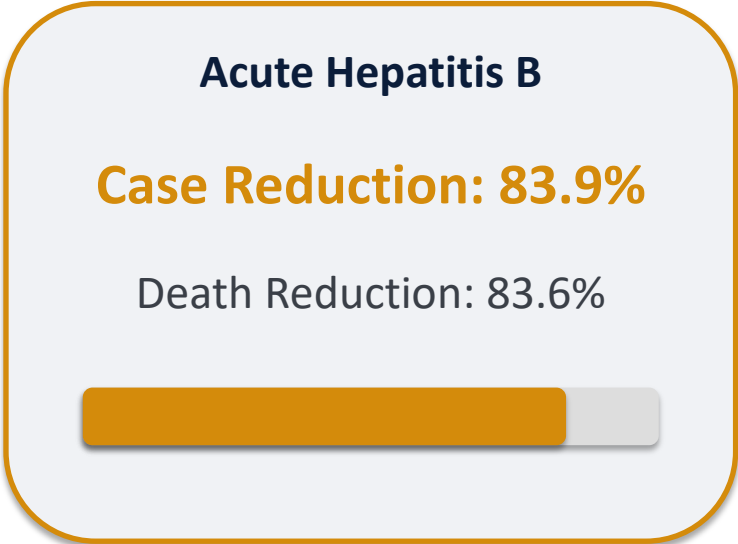
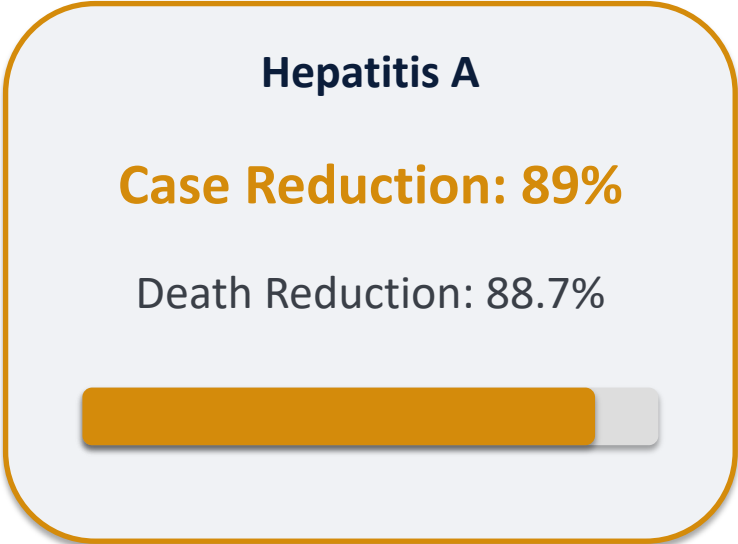
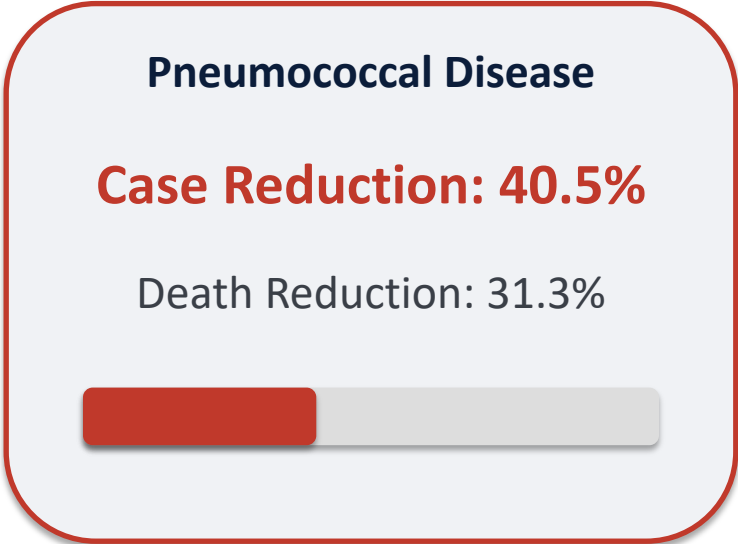
- The virus infected only humans — no animal reservoir
- Symptoms were easily distinguishable — enabling rapid identification
- The vaccine was cheap, effective, and heat-stable

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



Herd immunity shields newborns, immunocompromised individuals, and cancer patients on chemotherapy who cannot be vaccinated.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction



Barriers to Higher Coverage

Access Gaps
Not everyone globally has access to vaccines

Supply Shortages
Put children and vulnerable populations at risk

Trust & Understanding
Poor scientific literacy reduces uptake

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1 Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US.
- 2 Vaccine development timelines have collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.
- 3 Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.
- 4 Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign.
- 5 Gaps persist for pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access, coverage, and newer vaccines.

Call to Action: Strengthen global health systems — improve vaccine access, close coverage gaps, and rebuild public trust.